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Module 3 — Cloth Segmentation AI

In this module, I worked on Cloth Segmentation AI, where the objective was to identify and separate clothing regions from an image using AI segmentation techniques. Cloth segmentation helps distinguish different areas such as clothing, human body, and background at the pixel level.

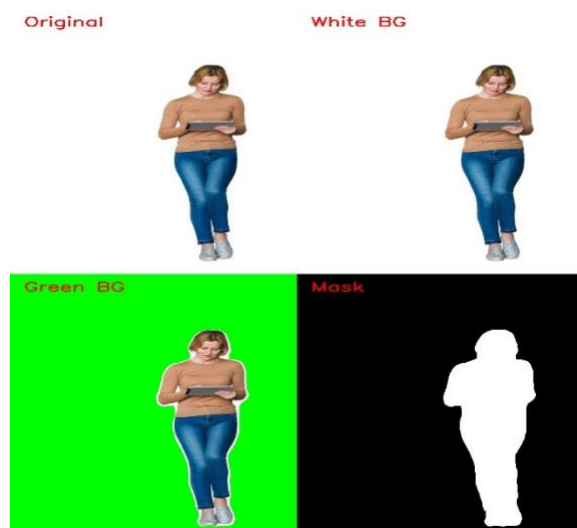
For this task, I explored and compared two segmentation models: Media Pipe Selfie Segmentation and DeepLabV3 to understand differences in segmentation quality, accuracy, and performance.

Model 1 -- Media Pipe

In this task, MediaPipe Selfie Segmentation was used to understand the segmentation workflow and test how effectively it separates a person from the background.

Observation:

MediaPipe produced fast and satisfactory results but showed less detail around clothing boundaries compared to advanced segmentation models.

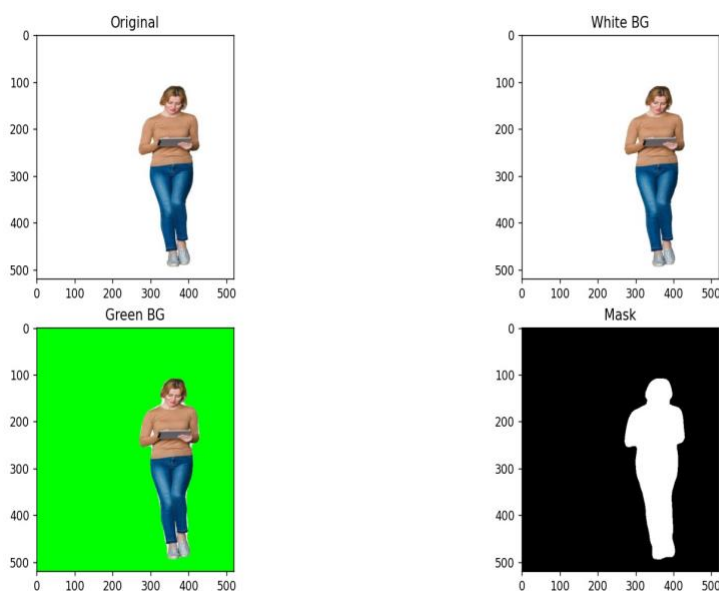


Model-2 -- DeepLabV3 Segmentation Testing

DeepLabV3 was used to perform detailed semantic segmentation and analyse segmentation quality. The input image was processed to generate a more precise segmentation mask with improved clothing and body region extraction.

Observation:

DeepLabV3 generated higher-quality segmentation outputs and showed better performance in detecting clothing regions and maintaining boundary accuracy.



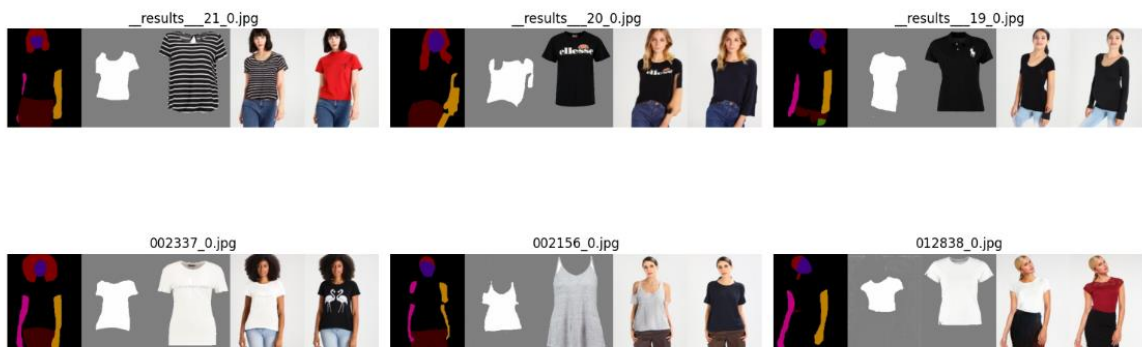
Study of DeepFashion Dataset

As part of this task, I explored the **DeepFashion dataset** to understand how fashion-related image data is prepared and used in segmentation models. The dataset includes images with clothing information that helps AI identify different clothing regions more effectively

```
input/
notebooks/
rkuo2000/
  deepfashion-tryon/
    __results__files/
    DeepFashion_Try_On/
    .idea/
      inspectionProfiles/
    ACGPN_inference/
      runs/
        G1G2/
      options/
        __pycache__/
      checkpoints/
        label2city/
      models/
        __pycache__/
        __pycache__/
      util/
        __pycache__/
      data/
        __pycache__/
      sample/
    images/
    Data_preprocessing/
      test_edge/
```

Sample Datasets

DeepFashion Dataset Samples

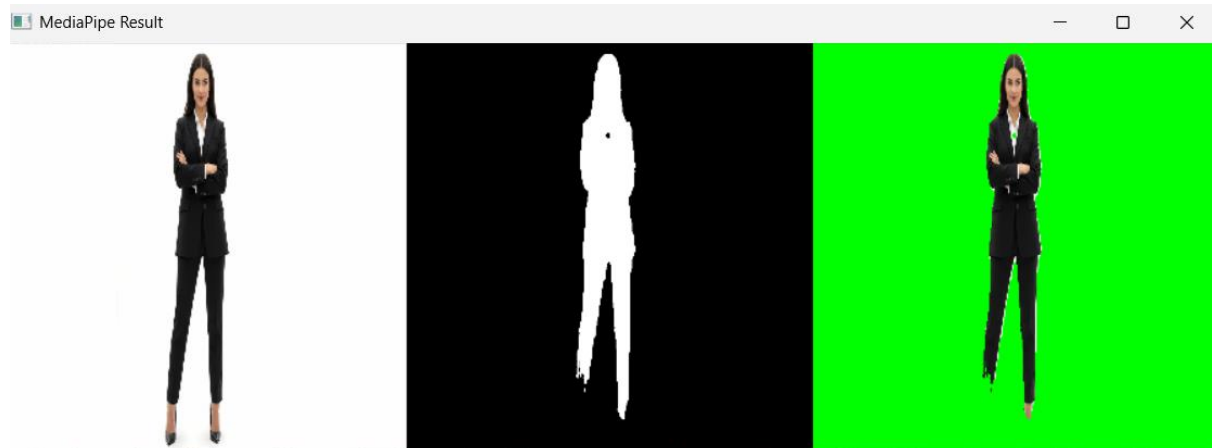


TESTING COMPLETE

MediaPipe Selfie Segmentation

Implemented MediaPipe Selfie Segmentation to understand the segmentation workflow and test how AI separates the foreground from the background. The model was provided with an input image and generated a segmentation mask to detect the human region.

```
C: > Users > Hansika > Downloads > mediapipe selfie > ...
1  import cv2
2  import mediapipe as mp
3  import numpy as np
4  # Load image
5  img = cv2.imread(r"C:\Users\Hansika\Downloads\Women.jpg")
6  rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
7  # MediaPipe Segmentation
8  with mp.solutions.selfie_segmentation.SelfieSegmentation(1) as seg:
9      result = seg.process(rgb)
10  mask = result.segmentation_mask > 0.5
11  green = np.zeros_like(img)
12  green[:] = (0, 255, 0)
13  output = np.where(mask[..., None], img, green)
14  mask_img = cv2.cvtColor(
15      (mask.astype(np.uint8) * 255),
16      cv2.COLOR_GRAY2BGR
17  )
18  final = np.hstack([
19      cv2.resize(img, (300,300)),
20      cv2.resize(mask_img, (300,300)),
21      cv2.resize(output, (300,300))
22  ])
23  cv2.imwrite("MediaPipe_Green_BG.jpg", final)
24  cv2.imshow("MediaPipe Result", final)
25  cv2.waitKey(0)
26  cv2.destroyAllWindows()
```



Module-4 Virtual Try – On AI

Virtual Try-On AI is a technology that digitally places clothes onto a person's image. The system first detects the body, then aligns the selected garment and adjusts it to match body position. Finally, it generates a virtual clothing output.

Workflow

Person Image



Select Garment



Detect Body



Align Cloth



Fit on Body



Generate Output

Research

Garment Warping

So garment warping is how AI takes a flat cloth image and makes it look like it's actually worn on a person. Like if you just paste a shirt on a photo it looks weird right ,that's where warping comes in.

It detects the body shape first Shoulders, chest, waist and then bends and stretches the cloth to fit around those points

So the steps are basically —

- Take the flat garment image

- Detect body key points from the person photo
- Warp the garment to match those points
- Output looks natural like the person is actually wearing It.

AI Fitting System – Short Research

AI Fitting System is a technology used in virtual try-on applications to estimate how clothes fit on a person without physically wearing them. It analyses body shape, pose, and measurements from images and adjusts the garment accordingly.

How it works:

- Detects human body and key points
- Estimates body size and fitting area
- Aligns and adjusts the garment on the body
- Generates a simulated clothing appearance

Simple Alignment Demo

Simple Alignment Demo is a basic virtual try-on technique used to place a garment image onto a human image by aligning it with body position. Instead of complex cloth simulation, this method uses body landmarks such as shoulders and torso to estimate where the clothing should appear.

Process:

- Load person image and garment image
- Detect body position using pose estimation
- Identify shoulder coordinates
- Resize garment based on body width
- Align and overlay garment onto the body
- Generate final try-on output

Original



Try-On Output

